Worksheet 3.1 — 3D Geometry: Volume, Surface Area and Distance

Mathematics: Analysis & Approaches SL (IB) · Unit 3: Geometry and Trigonometry

The Big Ideas

3D distance and volume formulas extend familiar 2D ideas one dimension further.

- Distance: $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$.
- Cone: $V = \frac{1}{3}\pi r^2 h$, curved area $\pi r l$. Sphere: $V = \frac{4}{3}\pi r^3$, area $4\pi r^2$.
- Angle between a diagonal and a face: find the base diagonal first, then use right-triangle trig with the height.

Worked Examples

Example 1 — 3D distance

Find the distance between $P(2,-1,4)$ and $Q(6,3,1)$.

Solution. $d = \sqrt{16+16+9} = \sqrt{41}$.

Conclusion: $d = \sqrt{41} \approx 6.40$.

Example 2 — Composite solid surface area

A cone (radius 5 m, height 12 m) is topped with a hemisphere of the same radius. Find the total exposed curved area.

Solution. Slant height $l = \sqrt{25+144} = 13$. Cone: $\pi(5)(13) = 65\pi$. Hemisphere: $\frac{1}{2}(4\pi \cdot 25) = 50\pi$.

Conclusion: total $= 115\pi \approx 361.3$ m^2.

Example 3 — Pyramid height

A square pyramid has base side 9 cm and volume 189 cm^3. Find its height.

Solution. Base area $= 81$; $189 = \frac{1}{3}(81)h = 27h$.

Conclusion: $h = 7$ cm.

Example 4 — Angle with the base

A box has length 12 cm, width 5 cm, height 9 cm. Find the angle the space diagonal makes with the base.

Solution. Base diagonal $= \sqrt{144+25} = 13$. $\tan\theta = \dfrac{9}{13} \Rightarrow \theta = \tan^{-1}(0.6923)$.

Conclusion: $\theta \approx 34.7$ř.

Example 5 — Midpoint in 3D

Find the midpoint of $A(3,-5,7)$ and $B(-1,9,1)$.

Solution. Average each coordinate: $\left(\dfrac{3-1}{2}, \dfrac{-5+9}{2}, \dfrac{7+1}{2}\right)$.

Conclusion: midpoint $= (1,2,4)$.

Example 6 — Composite solid volume

An ice-cream-cone shape has a cone of radius 3 cm, height 8 cm, topped with a hemisphere of the same radius. Find the total volume.

Solution. Cone: $\frac{1}{3}\pi(9)(8) = 24\pi$. Hemisphere: $\frac{1}{2}\left(\frac{4}{3}\pi(27)\right) = 18\pi$.

Conclusion: total volume $= 42\pi \approx 131.9$ cm^3.

Example 7 — Angle between a base diagonal and an edge

A box's base has length 6 cm and width 8 cm. Find the angle the base's diagonal makes with the 6 cm edge.

Solution. $\tan\theta = \dfrac{8}{6} = \dfrac{4}{3} \Rightarrow \theta = \tan^{-1}(1.333)$.

Conclusion: $\theta \approx 53.1$ř.

Example 8 — Solve for an unknown dimension

A box has length x, width $x+2$, height 5 cm, and volume 175 cm^3. Find x.

Solution. $5x(x+2) = 175 \Rightarrow x(x+2) = 35 \Rightarrow x^2 + 2x - 35 = 0 \Rightarrow (x-5)(x+7) = 0$.

Conclusion: $x = 5$ cm (rejecting the negative root).

Example 9 — Application — surface to paint

A cylindrical silo (radius 3 m, height 10 m) has a hemispherical roof of the same radius. Find the total area to paint, excluding the base (which sits on the ground).

Solution. Cylinder curved: $2\pi(3)(10) = 60\pi$. Hemisphere: $2\pi(9) = 18\pi$.

Conclusion: total $= 78\pi \approx 245.0$ m^2.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find the distance between $A(1, 2, -3)$ and $B(4, -2, 9)$.

Question 2

Find the volume of a cone with radius 6 cm and height 10 cm.

Question 3

A sphere has volume 972π cm^3. Find its radius.

Question 4

A box has length 10 cm, width 4 cm, height 3 cm. Find the length of its space diagonal.

Question 5

Find the angle the space diagonal of a cube with side 8 cm makes with the base.

Question 6

A cylinder (radius 4 m, height 15 m) is capped with a hemisphere of the same radius; find the total curved surface area.

Question 7

A square pyramid has volume 320 cm^3 and height 15 cm; find its base side length.

Question 8

Find the midpoint of $P(-4, 6, 2)$ and $Q(8, -2, 10)$.

Question 9

An ice-cream-cone shape has a cone of radius 6 cm, height 8 cm, topped with a hemisphere of the same radius. Find the total volume (coefficient of π).

Question 10

A box has length x, width $x+3$, height 4 cm, and volume 216 cm^3. Find x.

Question 11 — Challenge

A rectangular box has length 15 cm, width 8 cm, and its space diagonal is 20 cm. Find its height.

Question 12 — Challenge

A box has length 12, width 5, height 9 (as in Example 4). Find the angle the space diagonal makes with the vertical edge (using the relationship between this angle and the base angle).

Question 13 — Challenge

A cylindrical tank (radius 5 m, height 12 m) with a hemispherical dome (same radius) needs painting on the curved wall and dome only. Find the total area (coefficient of π).

Question Recap & Answer Key

Question Recap

1. Find the distance between $A(1, 2, -3)$ and $B(4, -2, 9)$.
2. Find the volume of a cone with radius 6 cm and height 10 cm.
3. A sphere has volume 972π cm^3. Find its radius.
4. A box has length 10 cm, width 4 cm, height 3 cm. Find the length of its space diagonal.
5. Find the angle the space diagonal of a cube with side 8 cm makes with the base.
6. A cylinder (radius 4 m, height 15 m) is capped with a hemisphere of the same radius; find the total curved surface area.
7. A square pyramid has volume 320 cm^3 and height 15 cm; find its base side length.
8. Find the midpoint of $P(-4, 6, 2)$ and $Q(8, -2, 10)$.
9. An ice-cream-cone shape has a cone of radius 6 cm, height 8 cm, topped with a hemisphere of the same radius. Find the total volume (coefficient of π).
10. A box has length x, width $x + 3$, height 4 cm, and volume 216 cm^3. Find x.
11. A rectangular box has length 15 cm, width 8 cm, and its space diagonal is 20 cm. Find its height.
12. A box has length 12, width 5, height 9 (as in Example 4). Find the angle the space diagonal makes with the vertical edge (using the relationship between this angle and the base angle).
13. A cylindrical tank (radius 5 m, height 12 m) with a hemispherical dome (same radius) needs painting on the curved wall and dome only. Find the total area (coefficient of π).

✓ Answer Key

1. $\sqrt{9 + 16 + 144} = \sqrt{169} = 13$
2. $V = \frac{1}{3}\pi(36)(10) = 120\pi \approx 376.99$ cm^3
3. $\frac{4}{3}\pi r^3 = 972\pi \Rightarrow r^3 = 729 \Rightarrow r = 9$ cm
4. $\sqrt{100 + 16 + 9} = \sqrt{125} = 5\sqrt{5} \approx 11.18$ cm
5. base diagonal $= 8\sqrt{2}$; $\tan\theta = \frac{8}{8\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \theta \approx 35.3$ř
6. cylinder curved: $2\pi(4)(15) = 120\pi$; hemisphere: $2\pi(16) = 32\pi$; total $= 152\pi \approx 477.5$ m^2
7. $320 = \frac{1}{3}s^2(15) = 5s^2 \Rightarrow s^2 = 64 \Rightarrow s = 8$ cm
8. midpoint $= (2, 2, 6)$
9. cone $= \frac{1}{3}\pi(36)(8) = 96\pi$; hemisphere $= \frac{1}{2}\left(\frac{4}{3}\pi(216)\right) = 144\pi$; total $= 240\pi$
10. $4x(x + 3) = 216 \Rightarrow x(x + 3) = 54 \Rightarrow x^2 + 3x - 54 = 0 \Rightarrow (x - 6)(x + 9) = 0 \Rightarrow x = 6$ cm
11. base diagonal $= \sqrt{225 + 64} = 17$; $h = \sqrt{400 - 289} = \sqrt{111} \approx 10.54$ cm
12. the angle with the vertical edge is the complement: 90ř $- 34.7$ř ≈ 55.3ř
13. cylinder curved: $2\pi(5)(12) = 120\pi$; hemisphere: $2\pi(25) = 50\pi$; total $= 170\pi \approx 534.1$ m^2